

MICHAEL PANASHE MUDIMBU

Senior Software Engineer & AI Systems Architect

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Portfolio: <https://michaelmudimbu.vercel.app> | **GitHub:** <https://github.com/micpana>

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer & AI Systems Architect with 7+ years of professional experience designing, building, deploying, and maintaining production-grade software systems, enterprise backend services, and AI-powered applications. Specialized in JavaScript, TypeScript, Python, Node.js, and PostgreSQL, with deep expertise in WhatsApp/Telegram chatbot technology, computer vision workflows, LLM integrations, and cloud infrastructure management. Proven track record of independently taking complex systems from architectural design to secure cloud deployment, proactive monitoring, and long-term production maintenance. Adept at bridging technical execution with strategic business requirements, ensuring rigorous compliance with security standards and performance optimization.

CORE TECHNICAL COMPETENCIES & SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, SQL, Bash Scripting.
- **Backend Architecture & Messaging:** Node.js, FastAPI, Flask, RESTful APIs, WhatsApp Business API, Telegram Bot API, WebSockets, Webhooks, Microservices.
- **Frontend Technologies:** React, React Native, HTML5, CSS3, Tailwind CSS, Responsive UI Design.
- **Databases & Data Management:** PostgreSQL, MongoDB, Supabase, Relational Database Design, Query Optimization, Data Pipelines.
- **AI, ML & Automation:** OpenAI API, LLM Integrations, WhatsApp/Telegram Chatbot Development, Computer Vision (OpenCV, MediaPipe), RAG, Dataset Engineering, Social Media Data Automation, Web Scraping & Browser Automation (Selenium, Playwright).
- **Cloud, DevOps & Infrastructure:** Linux (Ubuntu), Docker, Nginx Reverse Proxies, AWS, Git & GitHub Version Control, CI/CD Pipelines, GitHub Actions, Cloud Server Management & Security.
- **Development Practices & Governance:** Enterprise Software Architecture, Secure Coding, IT Security Best Practices, Code Reviews, Agile/Scrum Methodologies, Technical Estimation, Production Debugging & System Monitoring.

PROFESSIONAL EXPERIENCE

Nuvantica Software (Renovality Group) • Czech Republic (Remote)

Contract Fullstack Software Engineer (AI & Systems) | September 2025 - August 2026

- Served as the primary independent engineer responsible for designing, developing, deploying, and scaling AI-powered enterprise software systems and backend services.
- Architected and delivered Skinraya, a production AI skincare analysis platform, owning end-to-end development across the React frontend, backend API services, PostgreSQL database schema, and cloud infrastructure.
- Developed complex computer vision and LLM-powered workflows, integrating generative AI capabilities into production application functionality to automate user recommendations.
- Managed Linux production environments, configuring Nginx reverse proxies, Docker containerization, and cloud-hosted infrastructure with proactive security monitoring.
- Implemented proactive server monitoring and automated deployment practices focused on application reliability, uptime, and maintainability.
- Conducted rigorous technical reviews, enforced code quality standards, and collaborated with remote cross-functional stakeholders to deliver production-ready features.

Independent Fullstack Software Engineer • Freelance Client Projects (Worldwide)

Independent Software Architect & Developer | July 2023 - August 2025

- Designed and delivered production SaaS applications, AI-powered messaging bots (WhatsApp/Telegram), and business automation tools for international clients.
- Translated complex business requirements into robust technical specifications, system architectures, database designs, APIs, and implementation plans.
- Engineered scalable backend services using Python, FastAPI, Flask, and Node.js, paired with modern frontend architectures built in React and TypeScript.
- Implemented automated WhatsApp/Telegram chatbot workflows, third-party API integrations, and web scrapers to support business process interoperability.
- Configured Git-based development workflows, CI/CD deployment pipelines, and cloud infrastructure, maintaining 99.9% uptime across cloud-hosted Linux environments.
- Managed end-to-end technical execution, from initial effort estimation and backlog prioritization through development, testing, deployment, and ongoing production maintenance.

- Provided technical leadership and independent problem-solving for diverse client requirements, ensuring high-quality software delivery.

ESpace • Harare, Zimbabwe

Contract Fullstack Software Developer | February 2021 - July 2023

- Delivered custom full-stack software solutions and client-facing web and mobile applications for agency clients under strict Agile/Scrum delivery timelines.
- Designed and deployed robust web and mobile applications using React, React Native, Python, and backend APIs.
- Collaborated with Product Managers and remote stakeholders to review requirements, prioritize technical tasks, and ensure compliance with project specifications.
- Maintained Git version control workflows, branch management, and automated deployment scripts.
- Participated in technical problem-solving, code optimization, and application maintenance across diverse client projects.

Freelance Software Developer • Independent Client Projects

Software Developer | July 2019 - January 2021

- Developed bespoke websites, web applications, and business systems for local and international clients.
- Engineered RESTful APIs, relational database schemas, and responsive user interfaces using HTML, CSS, JavaScript, and Python.
- Managed software architecture, CI/CD pipelines, Git version control, and cloud infrastructure, ensuring 99.9% uptime and system security.
- Established practical experience delivering production software independently, from requirements gathering to hosting and maintenance.

Health Informatics Training and Research Advancement Centre (HITRAC) • Harare, Zimbabwe

Intern Fullstack Software Developer | August 2018 - July 2019

- Contributed to frontend and backend development for national health information systems, utilizing Java Spring Boot and React applications.
- Supported technical deployment and maintenance tasks for Zimbabwe's national Electronic Health Records (EHR) system (Impilo).
- Assisted in maintaining secure data pipelines and server environments compliant with national health informatics standards.

Omni Contact Centre • Harare, Zimbabwe

Customer Experience Associate & Sales Consultant | April 2026 - Present

- **Outbound Customer Experience Associate (April 2026 - June 2026):** Orchestrated and managed high-volume customer engagement campaigns, ensuring consistent service quality and operational efficiency under high-pressure performance targets. Resolved complex retention challenges through strategic negotiation and proactive problem-solving.
- **Global Sales Consultant (Diaspora Insurance - UK Financial Services Campaign) (July 2026 - Present):** Execute high-value outbound sales and client engagement campaigns for a UK-based financial services organization, ensuring strict data compliance, CRM accuracy, and professional communication standards.
- **Stakeholder Management:** Strengthened cross-functional communication and negotiation skills directly applicable to managing client-facing IT requirements and technical stakeholder expectations.

KEY VERIFIED PROJECTS & SYSTEMS BUILT

1. Skinraya - AI Skincare Analysis Platform

Production AI platform built for real-world commercial use (<https://app.skinraya.cz>)

- **Key Contributions:** Built and maintained the complete frontend, backend, and AI processing pipeline; implemented computer vision workflows for skin analysis; integrated AI-driven recommendation generation; designed and deployed production infrastructure.

- **Technologies:** React • TypeScript • FastAPI • PostgreSQL • OpenAI • Computer Vision • Docker • Nginx • Ubuntu Linux

2. Skin Segmentation Dataset Builder

Automated dataset engineering tool (<https://github.com/micpana/Skin-Segmentation-Dataset-Builder>)

- **Key Contributions:** Developed automated image processing workflows for dataset preparation; generated background-free segmentation datasets suitable for machine learning training pipelines; reduced manual dataset preparation effort through automated processing.

- **Technologies:** Python • OpenCV • MediaPipe • Computer Vision • Dataset Engineering

3. Herbal-AI

AI-powered recommendation platform (<https://github.com/micpana/Herbal-AI>)

- **Key Contributions:** Built AI-assisted recommendation workflows using modern LLM technologies; implemented backend services for user profiling and recommendation generation; designed API-driven architecture supporting scalable AI interactions.

- **Technologies:** React • FastAPI • Python • OpenAI API • LLM Integration • Prompt Engineering

4. CubePOS - Point of Sale & Business Management Platform

End-to-end POS and business management platform designed for retail operations.

- **Key Contributions:** Designed and developed frontend and backend systems; implemented inventory management, sales tracking, and reporting functionality; built scalable APIs and database architecture; managed deployment and production hosting.

- **Technologies:** React • Python • MongoDB • REST APIs • Docker

EDUCATION

Harare Institute of Technology • Harare, Zimbabwe

Bachelor of Technology Honours Degree in Information Technology

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **HP Life Foundation:** Agile Project Management | 2026
- **Elevify:** Data Annotation | 2026
- **The Forage:** Data Labeling Job Simulation | 2026
- **DataLense:** Essentials of Data Labeling & Annotation for AI Development | 2026
- **DataLense:** Human-in-the-Loop (HITL) QA & Data Quality Management | 2026